Conversii

Conversia nr. întregi (părţii întregi) prin împărţiri succesive (se recomandă să se utilizeze pornind de la o bază m. mare)

Calculele se efectuează în baza sursă

$$X_{(p)} = Y_{(q)} = r_m r_{m-1} \dots r_1 r_0{}_{(q)}$$

$$x_{(p)} : y_{(p)} = c_{0(p)} \text{ rest } r_{0(p)}$$

$$c_{0(p)} : 2_{(p)} = c_{1(p)} \text{ rest } r_{1(p)}$$

$$\vdots$$

$$c_{m-1(p)} : 2_{(p)} = c_{m(p)} \text{ rest } r_{m(p)}$$

$$\overset{\shortparallel}{0}$$

$542010_{(10)} = 1000112111110_{(3)}$

$542010 = 180670 = 60223 = 20074 = 6691 = 2230 =$
$= 743 = 247 = 82 = 27 = 9 = 3$
$= 1$

0111112110001

$1121_{(10)} = 461_{(16)}$

$1121 = 70 = 4$

$$\begin{array}{r} 16 \cdot \\ 7 \\ \hline 112 \end{array} \qquad \begin{array}{r} 16 \cdot \\ 4 \\ \hline 64 \end{array}$$

164

Conversia nr. subunitare (părții subunitare) prin înmulțiri succesive (se recomandă să se utilizeze pornind de la o bază m. mare)

Calculele se efectuează în baza sursă

$$0,x_{(p)} = 0,?_{(q)} = 0,y_{(q)} = 0,y_{-1}\ y_{-2} \dots {}_{(q)}$$

$$0,x_{(p)} * q_{(p)} = y_{-1},\ F_{1}{}_{(p)}$$

$$0,F_{1}{}_{(p)} * q_{(p)} = y_{-2},\ F_{2}{}_{(p)}$$

$$0,F_{2}{}_{(p)} * q_{(p)} = y_{-3},\ F_{3}{}_{(p)}$$

$$\vdots$$

condiția de oprire: $-F_n = 0$ (termină)

$-F_n = F_k,\ k < n$ - perioadă

- când s-au obținut suficiente cifre

$0,124_{(10)} = 0,1FB_{(16)}$ (cu 3 cifre după virgulă)

$$\begin{array}{r} 0,124\cdot \\ 16 \\ \hline 744 \\ 124 \\ \hline 1,984 \end{array} \qquad \begin{array}{r} 0,984\cdot \\ 16 \\ \hline 5904 \\ 984 \\ \hline 15,744 \end{array} \qquad \begin{array}{r} 0,744\cdot \\ 16 \\ \hline 4464 \\ 744 \\ \hline 11,904 \end{array}$$

3203110333133032 3)

$0,348_{(10)} = 0,(112101122013022300020030102123221 1$

$$\begin{array}{r} 0,348\cdot \\ 4 \\ \hline 1,392 \end{array} \qquad \begin{array}{r} 0,392\cdot \\ 4 \\ \hline 1,568 \end{array} \qquad \begin{array}{r} 0,568\cdot \\ 4 \\ \hline 2,272 \end{array} \qquad \begin{array}{r} 0,272\cdot \\ 4 \\ \hline 1,088 \end{array} \qquad \begin{array}{r} 0,088\cdot \\ 4 \\ \hline 0,352 \end{array}$$

$$\begin{array}{r} 0,352\cdot \\ 4 \\ \hline 1,408 \end{array} \qquad \begin{array}{r} 0,408\cdot \\ 4 \\ \hline 1,632 \end{array} \qquad \begin{array}{r} 0,632\cdot \\ 4 \\ \hline 2,528 \end{array} \qquad \begin{array}{r} 0,528\cdot \\ 4 \\ \hline 2,112 \end{array}$$

$0,112 \cdot 4 = 0,448$ | $0,448 \cdot 4 = 1,792$ | $0,792 \cdot 4 = 3,168$ | $0,168 \cdot 4 = 0,672$ | $0,672 \cdot 4 = 2,688$ | $0,688 \cdot 4 = 2,752$

$0,752 \cdot 4 = 3,008$ | $0,008 \cdot 4 = 0,032$ | $0,032 \cdot 4 = 0,128$ | $0,128 \cdot 4 = 0,512$ | $0,512 \cdot 4 = 2,048$ | $0,048 \cdot 4 = 0,192$

$0,192 \cdot 4 = 0,768$ | $0,768 \cdot 4 = 3,072$ | $0,072 \cdot 4 = 0,288$ | $0,288 \cdot 4 = 1,152$ | $0,152 \cdot 4 = 0,608$ | $0,608 \cdot 4 = 2,432$

$0,432 \cdot 4 = 1,728$ | $0,728 \cdot 4 = 2,912$ | $0,912 \cdot 4 = 3,648$ | $0,648 \cdot 4 = 2,592$ | $0,592 \cdot 4 = 2,368$ | $0,368 \cdot 4 = 1,472$

$0,472 \cdot 4 = 1,888$ | $0,888 \cdot 4 = 3,552$ | $0,552 \cdot 4 = 2,208$ | $0,208 \cdot 4 = 0,832$ | $0,832 \cdot 4 = 3,328$ | $0,328 \cdot 4 = 1,312$

$0,312 \cdot 4 = 1,248$ | $0,248 \cdot 4 = 0,992$ | $0,992 \cdot 4 = 3,968$ | $0,968 \cdot 4 = 3,872$ | $0,872 \cdot 4 = 3,488$ | $0,488 \cdot 4 = 1,952$

$0,952 \cdot 4 = 3,808$ | $0,808 \cdot 4 = 3,232$ | $0,232 \cdot 4 = 0,928$ | $0,928 \cdot 4 = 3,712$ | $0,712 \cdot 4 = 2,848$ | $0,848 \cdot 4 = 3,392$

Substituţia

$$a_n a_{n-1} \dots a_1 a_0, a_{-1} a_{-2} \dots a_{-m}(p) = a_n(q) * p^n(q) +$$
$$+ a_{n-1}(q) * p^{n-1} + \dots + a_0(q) \cdot p^0(q) + a_{-1}(q) \cdot p^{-1}(q) + \dots + a_{-m}(q) \cdot p^{-m}(q)$$

$$\overset{2}{2}\overset{1}{0}\overset{0}{5},\overset{-1}{0}\overset{-2}{1}_{(6)} = 2 \cdot 6^2 + 0 \cdot 6^1 + 5 \cdot 6^0 + 0 \cdot 6^{-1} + 1 \cdot 6^{-2} =$$
$$= 2 \cdot 36 + 5 + \frac{1}{36} =$$
$$= 72 + 5 + \frac{1}{36} = 77 + \frac{1}{36} = 77,02_{(10)}$$

$1 : 36 = 0,02$

0
10
0
100

16 ·
16
96
16
256

$7 : 16 = 0,43$

0
70
64
= 60
48
12

$A51,7_{(16)} = 10 \cdot 16^2 + 5 \cdot 16^1 + 1 \cdot 16^0 + 7 \cdot 16^{-1} =$

$= 2560 + 80 + \frac{7}{16} + 1$

$= 2641 + \frac{7}{16} = 2641,43_{(10)}$

Conversii rapide

ideea: nr. în baza $p \to p^k$:

$a_n \cdot p^n + a_{n-1} \cdot p^{n-1} + \dots + a_0 \cdot p^0 + a_{-1} \cdot p^{-1} + \dots + a_{-m} \cdot p^{-m}$

$\dots c_{k+1} \cdot p^1 + c_k \cdot p^0) p^{k^1} \cdot (a_{k-1} \cdot p^{k-1} + \dots + a_0 \cdot p^0) \cdot p^{k^0} + (a_{-1} \cdot p^{k-1} + \dots +$

$+ a_{-k}) \cdot (p^k)^{-1} + \dots$

2	8
000	0
001	1
010	2
011	3
100	4
101	5
110	6
111	7

2	16
0000	0
0001	1
0010	2
0011	3
0100	4
0101	5
0110	6
0111	7
1000	8
1001	9
1010	A
1011	B
1100	C
1101	D
1110	E
1111	F

$101\ 101\ 001,011\ 101\ 100\ 111\ 011_{(2)} = 551,35473_{(8)}$

$0010\ 0111\ 0101\ 1001,1101\ 0001\ 1001\ 1000\ 0101_{(2)} =$

$= 2759,D1985_{(16)}$

$3451,2210_{(8)} = 011\ 100\ 101\ 001,010\ 010\ 001\ 000_{(2)} =$

$= 0111\ 0010\ 1001,0100\ 1000\ 1000_{(2)} =$

$= 729,488_{(16)}$

$5BD2,3A_{(16)} = 0101\ 1011\ 1101\ 0010,0011\ 1010_{(2)} =$

$= 000\ 101\ 101\ 111\ 010\ 010,001\ 110\ 100_{(2)} =$

$= 055722,164_{(8)} = 55722,164_{(8)}$

Conversia prin bază intermediară